Cycle Test 2: worked solutions

17 Oct 2024 · 20 marks · 1 hour · CEPC12

Q1

3 marks

> What is metacentric height? Explain the conditions of equilibrium of a floating body. A metallic body floats at the interface of mercury and water so that 30% of its volume is in mercury and 70% in water. Find the density of the body.

Metacentric height (GM): the distance between the **centre of gravity G** of a floating body and its **metacentre M** (the point where the line of action of buoyancy in a slightly tilted position meets the original vertical axis). GM = BM − BG, with BM = I/V.

Conditions of equilibrium of a floating body:

- **Stable**: M above G (GM > 0). On tilting, weight and buoyancy form a **restoring couple**.
- **Unstable**: M below G (GM < 0). The couple **overturns** the body further.
- **Neutral**: M coincides with G (GM = 0). The body stays in any tilted position.

Numerical: weight of body = total buoyancy (from both liquids). Let V = volume of the body.

$\rho_b \, g \, V = (0.3V \times 13{,}600 + 0.7V \times 1000) \, g$

$\rho_b = 4080 + 700 =$ **4780 kg/m^3**

Q2

4 marks

> Differentiate between steady and unsteady flows. Show that the stream function $\psi = 6x - 4y + 7xy + 9$ represents an irrotational flow. Find its velocity potential.

Steady flow	Unsteady flow
Fluid properties (V, p, ρ) at a point **do not change with time**: $\partial V/\partial t = 0$	Properties at a point **change with time**: $\partial V/\partial t \neq 0$
Streamlines, pathlines and streaklines coincide	They generally differ
Constant discharge in a pipe	Flow through a pipe while a valve is being closed; draining tank

Irrotational check: a flow is irrotational if ψ satisfies Laplace's equation.

$\partial\psi/\partial x = 6 + 7y \rightarrow \partial^2\psi/\partial x^2 = 0$; $\partial\psi/\partial y = -4 + 7x \rightarrow \partial^2\psi/\partial y^2 = 0$

$\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 = 0 \rightarrow$ the flow is **irrotational**

Velocity potential ($u = \partial\psi/\partial y = \partial\phi/\partial x$; $v = -\partial\psi/\partial x = \partial\phi/\partial y$):

$u = 7x - 4$; $v = -(6 + 7y)$

$\partial\phi/\partial x = 7x - 4 \rightarrow \phi = 3.5x^2 - 4x + f(y)$

$\partial\phi/\partial y = f'(y) = -6 - 7y \rightarrow f(y) = -6y - 3.5y^2 + C$

$$\boxed{\phi = 3.5x^2 - 3.5y^2 - 4x - 6y + C}$$

(The other common sign convention, $u = -\partial\psi/\partial y = -\partial\phi/\partial x$, gives the same ϕ.)

Q3

4 marks

> Explain Euler's equation of motion. Water flows through a tapering pipe 100 m long with diameters 600 mm at the upper end and 300 mm at the lower end, at 50 l/s. The pipe has a slope of 1 in 30. Find the pressure at the lower end if the pressure at the upper end is 19.62 N/cm².

Euler's equation of motion is Newton's second law applied to a fluid element moving along a streamline, considering only **pressure and gravity** forces (viscous forces neglected). For an element of area dA and length ds:

$$-(\partial p/\partial s)\, ds\, dA - \rho g\, dA\, ds \cos\theta = \rho\, dA\, ds \times V\, (\partial V/\partial s)$$

With $\cos\theta = dz/ds$: $\mathbf{dp/\rho + g\, dz + V\, dV = 0}$. Integrating for an incompressible fluid gives **Bernoulli's equation** $p/\rho g + V^2/2g + z = \text{constant}$.

Numerical (lower end as datum; rise = 100/30 = 3.333 m):

$V_1 = 0.05/(\pi/4 \times 0.6^2) = 0.1768$ m/s; $V_2 = 0.05/(\pi/4 \times 0.3^2) = 0.7074$ m/s

$p_1/\rho g = 196{,}200/9810 = 20.0$ m

$p_2/\rho g = p_1/\rho g + V_1^2/2g + z_1 - V_2^2/2g = 20.0 + 0.0016 + 3.333 - 0.0255 = 23.309$ m

$p_2 = 23.309 \times 9810 =$ **$228{,}665$ N/m² = 22.87 N/cm²**

Q4

4 marks

> A window in a vertical wall is 12 m above the ground. A jet of water from a 50 mm nozzle is to strike the window. The flow through the nozzle is 40 l/s, and the nozzle is 1 m above the ground. Find the greatest horizontal distance of the nozzle from the wall so that the jet strikes the window.

Jet velocity $U = 0.04/(\pi/4 \times 0.05^2) = 20.37$ m/s; height to rise $y = 12 - 1 = 11$ m

Trajectory: $y = x\tan\theta - g x^2 (1 + \tan^2\theta)/(2U^2)$

For a given x this is a quadratic in tanθ; the **largest x** that can still reach y = 11 m is when the two roots coincide (discriminant = 0):

$x^2 = (U^2/g)^2 - 2y(U^2/g)$; $U^2/g = 20.37^2/9.81 = 42.30$ m

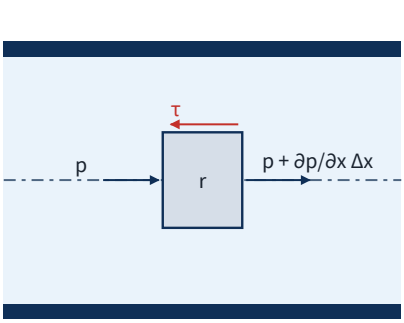
$x^2 = 42.30^2 - 2 \times 11 \times 42.30 = 1789.7 - 930.7 = 859.0$

$x =$ **29.31 m**, with jet angle $\tan\theta = (U^2/g)/x = 1.443 \rightarrow \theta = 55.3°$

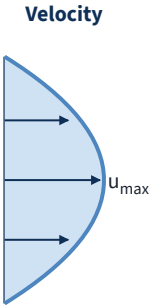
Q5

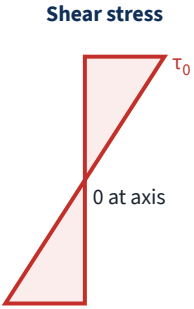
5 marks

> Derive the expressions for velocity distribution and shear stress distribution across a section for flow of a viscous fluid through a circular pipe. An oil of viscosity 10 poise flows between two parallel fixed plates 50 mm apart. Find the rate of flow if the drop of pressure in a length of 1.2 m is 0.3 N/cm². The width of the plates is 200 mm.



cylindrical element of radius r, length Δx





Shear stress. For a coaxial cylinder of radius r and length Δx, pressure force = shear force:

$$[p - (p + \partial p/\partial x\,\Delta x)]\,\pi r^2 = \tau\,(2\pi r\,\Delta x) \Longrightarrow \boxed{\tau = (-\partial p/\partial x)\,r/2}$$

So τ varies **linearly** with r: zero at the centre, maximum $\tau_0 = (-\partial p/\partial x)R/2$ at the wall.

Velocity. With $\tau = -\mu\, du/dr$: $du/dr = -(1/2\mu)(-\partial p/\partial x)\, r$. Integrating and using u = 0 at r = R:

$$\boxed{u = (1/4\mu)(-\partial p/\partial x)(R^2 - r^2)}$$

A **parabola**; $u_{max} = (1/4\mu)(-\partial p/\partial x)R^2$ at the centre; mean velocity $\bar{u} = u_{max}/2$; pressure drop $\Delta p = 32\mu\bar{u}L/D^2$ (Hagen–Poiseuille).

Numerical (fixed plates, laminar): μ = 10 poise = 1.0 N·s/m²; t = 0.05 m; Δp = 0.3 N/cm² = 3000 N/m² over L = 1.2 m; width b = 0.2 m.

For flow between fixed plates, $\Delta p = 12\,\mu\,\bar{u}\,L / t^2 \rightarrow \bar{u} = \Delta p\, t^2/(12\,\mu\,L)$

$\bar{u} = 3000 \times 0.05^2 / (12 \times 1.0 \times 1.2) = 0.521$ m/s ($u_{max} = 1.5\bar{u} = 0.781$ m/s)

$Q = \bar{u} \times (b \times t) = 0.521 \times 0.2 \times 0.05 =$ **$0.00521\ m^3/s = 5.21$ litres/s**